

Tips to use partial least squares structural equation modelling (PLS-SEM) in knowledge management

Gabriel Cepeda-Carrion, Juan-Gabriel Cegarra-Navarro and Valentina Cillo

Abstract

Purpose – Structural equation modelling (SEM) has been defined as the combination of latent variables and structural relationships. The partial least squares SEM (PLS-SEM) is used to estimate complex cause-effect relationship models with latent variables as the most salient research methods across a variety of disciplines, including knowledge management (KM). Following the path initiated by different domains in business research, this paper aims to examine how PLS-SEM has been applied in KM research, also providing some new guidelines how to improve PLS-SEM report analysis.

Design/methodology/approach – To ensure an objective way to analyse relevant works in the field of KM, this study conducted a systematic literature review of 63 publications in three SSCI-indexed and specific KM journals between 2015 and 2017.

Findings – Our results show that over the past three years, a significant amount of KM works has empirically used PLS-SEM. The findings also suggest that in light of recent developments of PLS-SEM reporting, some common misconceptions among KM researchers occurred mainly related to the reasons for using PLS-SEM, the purposes of PLS-SEM analysis, data characteristics, model characteristics and the evaluation of the structural models.

Originality/value – This study contributes to that vast KM literature by documenting the PLS-SEM-related problems and misconceptions. Therefore, it will shed light for better reports in PLS-SEM studies in the KM field.

Keywords Partial least squares (PLS), Knowledge management (KM), Structural equation modelling (SEM)

Paper type Literature review

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1. Introduction

In recent years there has been a growing interest in knowledge management (KM), which has resulted in an exponential growth of almost 50 per cent per year in KM publications (Bontis and Serenko, 2009). Gaining control over management structures plays an important role in the KM domain. It can be noted that control in KM is chiefly obtained measuring unobserved variables and the aggregation of the different indicators of the unobserved construct. In fact, recent publications from *Knowledge and Process Management (KPM)*, *Knowledge Management Research and Practice (KMRP)* and *Journal of Knowledge Management (JKM)* show how managerial issues affecting the performance of the entire organisation can only be operationalised in terms of unobserved variables (Cegarra-Navarro *et al.*, 2012; Cegarra-Navarro and Moya, 2005; Wang and Byrd, 2017).

Latent variables (from Latin: present participle of lateo (“lie hidden”), unlike observable variables, are not directly observed but are rather inferred from other directly observed and measured variables through a mathematical model. Regarding this, numerous methods and data analysis technique have been developed to address these issues using unobserved

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variables. One of them is partial least squares structural equation modelling (PLS-SEM), which is a type of structural equation modelling based on variance, also called on components, because it is the type of latent variable that it uses (Rigdon, 2014). The goal of PLS-SEM is to predict unobserved variables through the input values of observations (Hair *et al.*, 2014). This algorithm was developed by Wold (1974) to estimate structural equation models and it is considered another mean of accomplishing structural equations models (Hair *et al.*, 2013).

Structural equation models that aim to explain unobserved variables in terms of latent variables are called “latent variable models”. In this vein, Rigdon (2016) states that the main differences between these two kinds of SEM (i.e. covariance-based SEM, generally known as CB-SEM, vs PLS-SEM) is the type of used latent variables. Recent advances in PLS-SEM reporting studies suggest that latent variables can be operationalised as either composites or factors (Benitez *et al.*, 2016). While the factor model hypothesises that the variance of a set of indicators can be perfectly explained by the existence of one unobserved variable, the composite model does not impose any restrictions on covariances between indicators of the same construct (Henseler, 2017). However, it should be noted here that although composites and factors are two different concepts that both require an adequate articulation and distinction, they were often used interchangeably throughout the published KM literature (Dijkstra and Henseler, 2015; Henseler *et al.*, 2016).

It should also be noted that many of the constructs used in the design of KM research are the result of human crafting. For instance, information about organisational culture, absorptive capacity, knowledge corridors, unlearning, organisational learning, just to mention a few of them, are obtained from managers and employees interviews. This can mean that many models used to operationalise KM constructs can be addressed through composites, which use linear combinations of indicator variables as proxies of the conceptual variables under investigation to explain the variance of the unobserved variables in the structural model (Rigdon *et al.*, 2017). In other words, due to the artificial and latent nature of KM variables (e.g. organisational learning, organisational agility or unlearning), these variables are likely to be composites rather than factors (Henseler, 2017). This means that PLS-SEM is a useful tool for KM researchers as it provides researchers with a seamless and secured method to process and analyse data collected in terms of data requirements, model complexity and relationships specification (Sarstedt *et al.*, 2014). Based on these arguments, many works published using PLS-SEM are inadequately substantiated or qualitatively unsatisfactory (Ali *et al.*, 2017). It can also be noted here that in the case of KM, methodological research in PLS-SEM should be detailed and substantive, not merely formal.

Following the path initiated by different domains in business research (e.g. Human Resource Management, Hospitality, Accounting, International Business, Tourism, by citing the most recent), this paper examines how PLS-SEM has been applied in KM research, and also provides some new guidelines how to report PLS-SEM analysis. Thus, this paper helps KM researchers to report their PLS-SEM studies. As methodological research in PLS-SEM is very dynamic in terms of evaluating the method's aspects and developing tools that increase its usefulness for applied researchers (Hair *et al.*, 2018; Richter *et al.*, 2016), it is especially important to understand the method's properties.

Based on a systematic review of current PLS-SEM applications in three Social Sciences Citation indexed (SSCI) KM journals between 2015 and 2017, updated guidelines for PLS-SEM use in KM research will be provided (Hair *et al.*, 2018). In the following sections, we will go through fundamental studies and themes within PLS-SEM research in KM issues, all of which are at the core of the debate. The paper is organised as follows; we discuss the potential of application of PLS-SEM in KM field. Next, we provide a review about how the application and reporting of PLS-SEM studies on KM have progressed in the past three years, which permits us to highlight main common mistakes to avoid. In Section 4, we

provide a few recommendations based on more common mistakes for KM researchers. And finally, in the conclusion section, we assert that KM researchers must continuously update and adapt their approach to PLS-SEM advancements to increase the research effort and exploit research results at best.

2. Partial least squares structural equation modelling studies and themes in knowledge management

2.1 Reasons for using partial least squares structural equation modelling

In recent years, SEM have expanded with the development of computer hardware and software. SEM belongs to multivariate-analysis family, where multiple variables are analysed. These variables in KM research are associated to firms, managers, customers, activities, and so forth. They are also obtained mainly directly from surveys and interviews, or indirectly through secondary data from existing databases. SEM also is considered a second-generation data analysis technique and has been used by social science researchers in past 30 years. SEM takes advantage of latent variables measured by indicator variables. There are two types of SEM: covariance-based SEM (CB-SEM) and partial least squares SEM (PLS-SEM). The first uses common factor model to represent latent variables and, however, PLS-SEM uses composites to model latent variables. Recently, [Rigdon \(2016\)](#) have incorporated this distinction between latent variables representing the unobserved conceptual variables. Therefore, as [Rigdon \(2016\)](#) states, we can consider three types of variables in a SEM model: unobserved conceptual variables, the observed variables and the proxies variables, operationalised either common factor or composites depending on several circumstances (i.e. research objectives, proxies' nature, researchers justification, [...]). In this paper, we focus on PLS-SEM as it emerged later and has been a slow development from its birth in the early 80s on the past century ([Hair et al., 2017](#)).

Much has been written about PLS-SEM since [Chin \(1998\)](#) published a nontechnical introduction to the PLS approach. In this vein, [Rigdon \(2016\)](#) identifies the following reasons that should not be used as a pretext to use the PLS approach:

- small sample size;
- non-normal data distribution;
- existence of formative indicators; and
- statement of exploratory research.

More recently, [Ali et al. \(2017\)](#), [Rigdon \(2016\)](#) and [Rigdon et al. \(2017\)](#) have highlighted that researchers use a technique that is consistent with the type of model that they intend to estimate. These ideas could mean that it is very important for researcher to specify the type of model they use because it determines the more appropriate data analysis technique ([Ali et al., 2017](#); [Rigdon et al., 2017](#)).

More generally, PLS-SEM researchers agree that those models who only use common factor models, should mainly consider Covariance-based Structural Equation Model (CB-SEM). While researchers using models with composites should use methods base on composites such as PLS-SEM ([Dijkstra and Henseler, 2011](#)), in the case of mixed models where both factors and composites are to be found, the option is limited to PLS consistent (PLSc) ([Dijkstra and Henseler, 2015](#)). On the basis of the considerations above, it is reasonable to assume that the first reason for using PLS-SEM is attributed to the type of latent variables in the path model that researchers intend to estimate (i.e. these models are intended primarily as speculative proposals and as conceptual constructs).

According to [Rigdon \(2012\)](#), the specified constructs in a path model are just proxies, and there is always a validity gap between these proxies and the true value of the theoretical

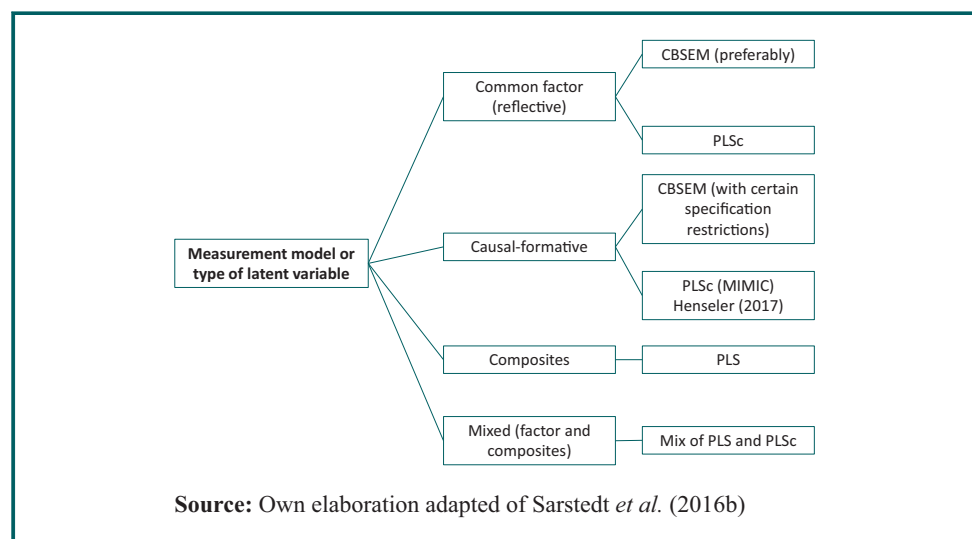
concept. The last ones are the main goal of path models' specifications. Thus, the specification of measures (i.e. latent variables) are named as measurement model in the SEM universe. Following [Henseler, Ringle and Sarstedt's \(2016\)](#) suggestions, we identify the following three kinds of measurement models:

1. common factor models (i.e. reflective measurement);
2. composite model; and
3. causal indicators model (i.e. formative measurement).

Another reason for using PLS-SEM is related to the nature of the construct. Although the type of latent variable or measurement model is determined by the researcher, [Henseler \(2017\)](#) suggests that the selection should be based on the nature of the construct. And by doing so, "behavioural constructs" and "design constructs" can be identified. On the one hand, when the construct independently exists in the reality as ontological entity associated to psychological characteristics (such as consumers' attributes and/or attitudes), it can be stated that this construct is "behavioural". Therefore, this construct is probably modelled as a common factor. If this construct possesses causal and formative indicators, it can be considered as "formative construct". On the other hand, when the construct is theoretical thought product, created by and for the research, and it is an artefact composed by elemental components like a mix of them, we can consider this construct as a "design construct", and, therefore, this is better modelled by means of composites ([Henseler et al., 2014; Henseler, 2017](#)). Following to [Henseler et al. \(2016\)](#) PLS-SEM can only estimates by composites. The implications of this election, as commented, determines the more suitable types of SEM (i.e. CB-SEM vs PLS-SEM). The works of [Rigdon et al. \(2017\)](#) and [Henseler \(2017\)](#) can help in this discussion.

Based on the work of [Sarstedt et al. \(2016a\)](#), [Figure 1](#) shows a guide for the election of the SEM approach depending on the type of latent variables being measured. Following the decision tree about data analysis technique according to the type of latent variable, researchers should use PLS-SEM for composites and CB-SEM (or PLSc) for (common) factors. Therefore, the remaining question to answer is, does the composite fairly represent the conceptual variable which one is attempting to include in the statistical model? While a composite reflective (mode A) construct shows a high level of correlation between

Figure 1 The election of the SEM approach



indicators (Dijkstra and Henseler, 2015; Henseler *et al.*, 2016), a composite formative construct (mode B) shows that its indicators are the ingredients to be combined to shape the construct itself (Benitez *et al.*, 2016).

Despite above positive views for using composites, there are also critical voices. For instance, Rönkkö and Evermann (2013) suggest that PLS does not in fact use latent variables. As they explain, PLS does not estimate path models with latent variables, but with composites. In this vein, Rigdon (2016) suggests that PLS-SEM is a kind of SEM because it uses a type of latent variable which is modelled through a network of sequential relationships (i.e. path model). The author also takes this opportunity to clarify the measurement error limitations established by Rönkkö and Evermann (2013). In this vein, he states that PLS-SEM does not work in the simulations contingencies used by Rönkkö and Evermann (2013) and therefore they “broke the (own) statistical method”(i.e. PLS-SEM) (Rigdon, 2016, p. 602).

Among the practical problems of PLS-SEM, Rönkkö and Evermann also concluded that “even simple summed scales provide better reliability than PLS, when used with regression, these traditional methods to generate composites have test statistics with known distributions allowing Null Hypothesis Significance Testing (NHST)” (Rönkkö and Evermann, 2013, p. 443). Again, Rigdon *et al.* (2017) challenge this issue using simulations for both factor based populations and composites simulations. These findings are supported by Sarstedt *et al.* (2016b), who emphasise that PLS-SEM offers more consistent coefficient estimates (i.e. less biased) for composites populations than other types of SEM (CB-SEM) and statistical methods.

Another interesting issue that has been criticised by some is due to the sample size requirements. As Goodhue *et al.* (2012) noted, the sample size requirements for PLS-SEM are generally larger than what some researchers assume. In this vein, while the use of PLS-SEM for very small sample sizes may be true for models containing only strong path coefficients and large effect sizes, it is not true for models with path coefficients of more modest magnitudes, and certainly not true for models with fairly weak path coefficients (Kock and Hadaya, 2018). In essence critical voices assert that when KM researchers are testing hypotheses, they may prefer to avoid the use of PLS.

In defence of PLS, it should be noted that PLS-SEM literature has reviewed constantly the software used to perform PLS-SEM. In fact, many of the above problems have been addressed with new PLS-SEM software packages. For example, new packages provide fit indexes and also allow researchers to operationalise unobserved variables as either composites or factors. In other words, many of the disadvantages have been properly solved over the past years with these latest packages and the increasing theoretical refinements (Henseler *et al.*, 2016). Bearing above ideas in mind, Table I summarises the key reasons that are referred to use PLS-SEM currently.

2.2 Review of partial least squares structural equation modelling applications in knowledge management

To ensure an objective way of analysing relevant works in the field of KM, this study conducted a systematic literature review approach (Jesson *et al.*, 2011; Tranfield *et al.*, 2003). First, a number of articles were identified for preselection. In doing so, we conducted a preliminary key word search with various combinations of buzzwords – search terms were “knowledge management AND structural equation,” “organisational learning AND structural equation,” “innovation AND structural equation,” We used several databases such as EBSCOhost, ABI/INFORM, EconBiz, ScienceDirect, Emerald Insight, JStor, SAGE Journals, and Google Scholar. These keywords were only searched for the title and/or abstract of studies. Hence, irrelevant works that only (casually) mention structural equation in the text or out of context were eliminated. We excluded false positives that only refer to the

Table I Key reasons for using PLS-SEM

Reasons	Source
Latent variables in path model are composites	Rigdon <i>et al.</i> (2017), Sarstedt <i>et al.</i> (2016a, 2016b)
Research objective is prediction of critical constructs	Cepeda <i>et al.</i> (2016), Shmueli <i>et al.</i> (2016)
Identify drivers constructs	Hair <i>et al.</i> (2014)
Secondary or file data are used	Gefen <i>et al.</i> (2011), Rigdon (2012)
Sample size is small because population is small	Richter <i>et al.</i> (2016)
Complex path models (i.e. high indicators numbers and constructs) are estimated	Chin (2010), Hair <i>et al.</i> (2017)
An exploratory approach is adopted in situation where there are abundant data, low theoretical support and no clear guide to specify the measurement models	Rigdon <i>et al.</i> (2017)
Latent variables scores are required, for example, high order constructs	Chin (2010)
Source: Own elaboration	

methodology in the text, without using PLS-SEM. Moreover, doublings were removed from the analysis. Furthermore, selected articles should have been published in the past three years at *KPM*, *KMRP* or *JKM*. Works that failed to fulfil this criterion were also eliminated from further examination.

Figure 2 shows the evolution of PLS-SEM works published every year that conform to the criterion mentioned above. After applying above criteria, a final number of 63 ($n = 63$) relevant works were identified ($KPM = 10$, $KMRP = 10$ and $JKM = 43$), which in turn reveals that a significant amount of KM works has empirically used PLS-SEM. Figure 2 also shows that *JKM* has been the KM journal which has published more PLS-SEM works during the past three years.

Table II depicts the works in KM journals included in the review process. These works significantly deal with different topics of KM. As Table II shows, the most common concepts include constructs such as those associated with innovation, social networks, intellectual capital, learning and knowledge in various forms and on different levels. Most of the authors of these papers acknowledge the difficulty and the limited amount of information to operationalise these measures (Huang and Chen, 2017; Lowik *et al.*, 2017). As Cegarra-Navarro *et al.* (2017) noted, the difficulty of taking into account these complex factors is due to their multi-dimensional aspects, the ambiguous impact of success factors, and vague

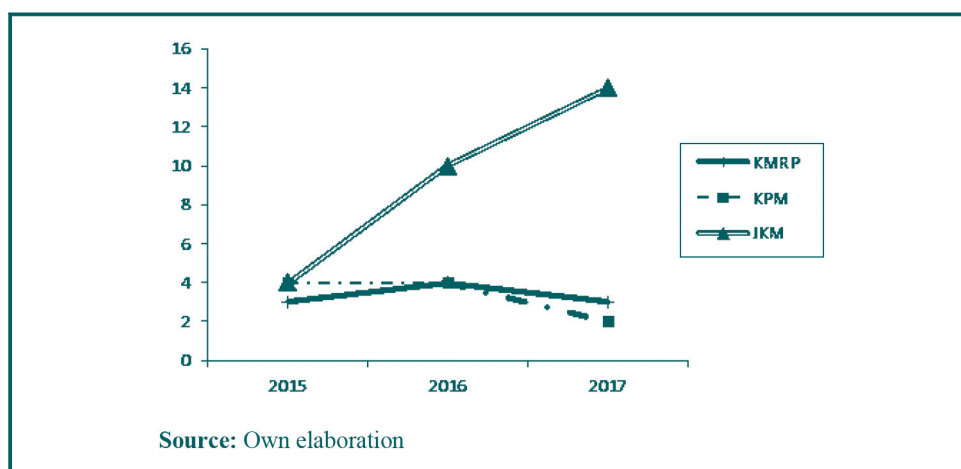
Figure 2 The evolution of PLS-SEM works

Table II KM works included in the review process

Source	Topic/constructs	Journal
Matić <i>et al.</i> (2017)	Organisational climate, motivational drivers, and empowering leadership	KMRP
Fidel <i>et al.</i> (2016)	Effects on innovation capacity and marketing results	KMRP
Park <i>et al.</i> (2017)	Social network and knowledge characteristics on job performance	KMRP
Vallat <i>et al.</i> (2016)	Double-loop learning	KMRP
Roxas and Chadee (2016)	Environmental sustainability	KMRP
Buenechea-Elberdin <i>et al.</i> (2017)	Human capital, renewal capital and entrepreneurial capital in innovation	KMRP
van Reijns <i>et al.</i> (2015)	Social capital on dynamic capability in organisations	KMRP
Jardon (2015)	Intellectual capital to obtain competitive advantages	KMRP
Hau <i>et al.</i> (2016)	Sharing tacit knowledge in practice	KMRP
Presbitero <i>et al.</i> (2015)	Intra- and inter-team dynamics on organisational learning: role of knowledge-sharing capability	KMRP
Demartini (2015)	Relationships between Social and Intellectual Capital:	KPM
Nabelsi <i>et al.</i> (2017)	Increasing the Impact of Wikis on Project Performance	KPM
Vieira <i>et al.</i> (2015)	Absorptive Capacity and Technology Knowledge: Enhancing Relational Capital	KPM
Mandal (2015)	Supply and Demand Effects on Supply Chain Flexibility	KPM
Wipawayangkool and Teng (2016)	Assessing Tacit Knowledge and Sharing Intention	KPM
Vargas and Lloria (2017)	Performance and Intellectual Capital	KPM
Costa and Monteiro (2016)	Knowledge Processes, Absorptive Capacity and Innovation	KPM
Olander <i>et al.</i> (2015)	HR-related Knowledge Protection and Innovation Performance	KPM
Raymond <i>et al.</i> (2016)	IT-enabled KM	KPM
Vanhala <i>et al.</i> (2016)	Organisational Trust Dimensions as Antecedents of Organisational Commitment	KPM
Ashok <i>et al.</i> (2016)	Investments in KM and innovation in services	JKM
Campos-Climent and Sanchis-Palacio (2017)	Absorptive capacity on shared value creation in social enterprises	JKM
Chhim <i>et al.</i> (2017)	Knowledge reuse through electronic knowledge repositories: a multi theoretical study	JKM
Sedighi <i>et al.</i> (2016)	Exploration of multi-layered knowledge sharing participation: the roles of perceived benefits and costs	JKM
Khedhaouria and Jamal (2015)	Sourcing knowledge for innovation: knowledge reuse and creation in project teams	JKM
Huang and Chen (2017)	Exploring the antecedents and consequences of the transactive memory system: an empirical analysis	JKM
Vătămănescu <i>et al.</i> (2016)	Harnessing network-based intellectual capital in online academic networks. From the organisational policies and practices towards competitiveness	JKM
Bharati <i>et al.</i> (2015)	Better knowledge with social media? Exploring the roles of social capital and organisational KM	JKM
Doloreux and Turkina (2017)	Are winemaker consultants just another source of knowledge for innovation?	JKM
Kim and Kim (2017)	The effect of compliance knowledge and compliance support systems on information security compliance behaviour	JKM
Akbari and Ghaffari (2017)	Verifying relationship of KM initiatives and the empowerment of human resources	JKM
Sharma <i>et al.</i> (2016)	Effects of KM on client-vendor relationship quality: the mediating role of global mindset	JKM
Xiao <i>et al.</i> (2017)	How does individuals' exchange orientation moderate the relationship between transformational leadership and knowledge sharing?	JKM
Lowik <i>et al.</i> (2017)	Antecedents and effects of individual absorptive capacity: a micro-foundational perspective on open innovation	JKM
Leal-Millán <i>et al.</i> (2016)	IT and relationship learning in networks as drivers of green innovation and customer capital: evidence from the automobile sector	JKM
Martin-Perez and Martin-Cruz (2015)	The mediating role of affective commitment in the rewards-knowledge transfer relation	JKM
Wang <i>et al.</i> (2015)	Encouraging knowledge contribution in IT support: social context and the differential effects of motivation type	JKM
Singh <i>et al.</i> (2017)	Factors affecting Web 2.0 adoption: exploring the knowledge sharing and knowledge seeking aspects in health care professionals	JKM
Serenko and Bontis (2016b)	Understanding counterproductive knowledge behaviour: antecedents and consequences of intra-organisational knowledge hiding	JKM
Giampaoli <i>et al.</i> (2017)	KM, problem solving and performance in top Italian firms	JKM

(continued)

Table II

Source	Topic/constructs	Journal
Cegarra-Navarro <i>et al.</i> (2017)	Linking procedural memory with organisational learning through knowledge corridors	JKM
Suh and Wagner (2017)	How gamification of an enterprise collaboration system increases knowledge contribution: an affordance approach	JKM
Lee <i>et al.</i> (2017)	Setting a knowledge boundary across teams: knowledge protection regulation for inter-team coordination and team performance	JKM
Kianto <i>et al.</i> (2016)	The impact of KM on job satisfaction	JKM
Lowik <i>et al.</i> (2016)	The team absorptive capacity triad: a configurational study of individual, enabling, and motivating factors	JKM
Serenko and Bontis (2016a)	Negotiate, reciprocate, or cooperate? The impact of exchange modes on inter-employee knowledge sharing	JKM
Lin and Lo (2015)	Mechanisms to motivate knowledge sharing: integrating the reward systems and social network perspectives	JKM
Wang and Byrd (2017b)	Business analytics-enabled decision-making effectiveness through knowledge absorptive capacity in health care	JKM
Ortiz <i>et al.</i> (2018)	Inter-organisational social capital as an antecedent of a firm's knowledge identification capability and external knowledge acquisition	JKM
Pee (2018)	Community's knowledge need and knowledge sharing in Wikipedia	JKM
Ali <i>et al.</i> (2017)	Impact of knowledge sharing and absorptive capacity on project performance: the moderating role of social processes	JKM
García-Villaverde <i>et al.</i> (2018)	The role of knowledge absorptive capacity on the relationship between cognitive social capital and entrepreneurial orientation	JKM
Albort-Morant <i>et al.</i> (2018)	Absorptive capacity and relationship learning mechanisms as complementary drivers of green innovation performance	JKM
Mirzaee and Ghaffari (2018)	Investigating the impact of information systems on knowledge sharing	JKM
Aboelmaged (2018)	Knowledge sharing through enterprise social network (ESN) systems: motivational drivers and their impact on employees' productivity	JKM
Lekhawipat <i>et al.</i> (2018)	How internal attributions affect knowledge sharing behaviour	JKM
Ritala <i>et al.</i> (2018)	External knowledge sharing and radical innovation: the downsides of uncontrolled openness	JKM
Claver-Cortés <i>et al.</i> (2018)	Strategic KM in subsidiaries and MNC performance. The role of the relational context	JKM
Rua <i>et al.</i> (2018)	Key drivers of SMEs export performance: the mediating effect of competitive advantage	JKM
Cabrilo and Dahms (2018)	How strategic KM drives intellectual capital to superior innovation and market performance	JKM
Source: Own elaboration		

and general nature of such factors. One possible explanation this study offers for this is the fact that KM concepts can realistically only be thought of as a proxy for an underlying latent phenomenon that is not fully measurable itself. As shown in [Table II](#), the research literature has operationalised all these constructs as precisely as possible by drawing on relevant conceptual frameworks and validated these models, to some extent, with practitioners.

The literature also shows many overlapping and sometimes conflicting lists of constructs (e.g. knowledge sharing, knowledge seeking or assessing knowledge). A possible explanation for these findings may relate to the fact that KM research field has not yet reached its maturity, which makes it a potential topic to conduct new quantitative research in this area. Another thing that seems to be common to the most common concepts of KM is that as they are simply metaphors which help researchers describe certain phenomena ([Tsang, 2017](#)), researchers “cannot measure the construct itself”. For example, creating an inventory of capacities or shared values, knowledge beliefs, assumptions, knowledge and knowledge structures at time t1 and comparing it to an inventory at time t0 to identify actual

deletions, modifications and additions is likely beyond the resources or capability of researchers though possible in principle.

The increasing growth of studies in KM, likewise, has shown the importance of developing efficient and simple programmes of being used for both researchers and practitioners. PLS-SEM literature has reviewed constantly the software used to perform PLS-SEM. The PLS-SEM software starts with the classic Jan-Bernd Lohnöller for FORTRAN called LVPLS1.8 (Lohmöller, 1989). In the eighties and nineties Wynne W. Chin developed the star product, it was the PLS-Graph's time (Chin *et al.*, 2003). From 2000s several software packages have appeared (e.g. VisualPLS, SmartPLS, XLSTAT-PLSPM, WrapPLS, and ADANCO). At the same time R packages arrived such as: PLSPM, SEMPLS and recently SEMinR. This basically is the full picture of the current software to perform PLS-SEM analysis.

Although many scholars have elaborated diverse guidelines to report PLS-SEM (Benitez *et al.*, 2016; Chin, 1998; Henseler *et al.*, 2016; Sarstedt *et al.*, 2016b), these guidelines need to be updated and adapted to the different specific domains (i.e. KM). Additionally, it has been considered in the last times a lot of development and controversies in PLS-SEM domain. These reviewed aspects help to make some brief guidelines to recognise and elaborate nice PLS-SEM reports in KM papers.

3. Partial least squares structural equation modelling recommendations for reporting in knowledge management

Once reviewing the PLS-SEM application on KM, we want to establish some minimum guidelines to report PLS-SEM analyses in KM. In fact, it is important for that the sound PLS-SEM application is reported adequately. As several PLS-SEM guidelines have been reported in the recent literature (Benitez *et al.*, 2016; Chin, 1998; Henseler *et al.*, 2016; Sarstedt *et al.*, 2016b), it has not been the goal for this paper developing a new guideline. This study proposes a few aspects that are often missed mainly in the PLS-SEM report and sometimes in the own PLS-SEM analysis. In doing so, we are going to focus on nine aspects to be considered on PLS-SEM report. They are:

1. reasons for using PLS-SEM;
2. purpose of PLS-SEM analysis;
3. types of latent variable and/or epistemological nature of the measure: measures and model specification;
4. assessment of global model fit;
5. assessment of measurement model;
6. assessment of structural model;
7. use of predictive analysis;
8. use of mediation analysis; and
9. the software package.

We explain each of these aspects below.

3.1 Reasons for using partial least squares structural equation modelling in knowledge management

After reviewing the literature shown in Table II, we have found evidence that there are many KM researchers who have used bad reasons to support PLS-SEM use. Regarding this, there were 56 KM studies reporting reasons for using PLS-SEM (88.8 per cent). Among the main reasons reported were: small or reduced sample size, 26 papers of 56 (46.42 per

cent); no distribution or non-normal distribution, 8 papers (32.14 per cent); scarce, new, or development of theory, 17 (39 per cent); complex models, 11 (19.64 per cent); prediction of explained variance of dependent variable, 10 (17.8 per cent). Other reasons were: using formative and reflective construct; testing causality; single case data; exploratory research; and composites and component models.

Based on the work of [Rigdon \(2016\)](#), the arguments of sample size and data distribution are not completely valid to justify using PLS-SEM. Thus, with respect to sample size, [Rigdon \(2016\)](#) states:

[...] the nature of the population that justifies the small sample size, and not the small sample size that justifies the choice of PLS path modeling (p. 600).

According to data distribution, PLS-SEM assume nonspecific data distribution, as normality for indicators and no independence of observations ([Chin, 2010](#)). But, it does not mean no data distribution works in PLS-SEM.

Likewise, statements as exploratory research when authors do not make exploratory research, are not valid too. The same works for researchers who declare “predictions purposes”. Therefore, KM researchers need to justify the use of PLS-SEM with arguments such as the nature of latent variables (composites or components), the complexity of models, and the need of using latent variables scores.

In fact, a recent study of [Sarstedt et al \(2016a, 2016b\)](#) shows as PLS-SEM exhibits no bias using composites and a small bias with common factor models. In turn, for CB-SEM, no bias is found for common factor models and bigger bias for composites. Additionally, due to PLS-SEM algorithm segments the model using different set of regressions, the complexity of the model is not an issue when PLS-SEM is estimated. Finally, due to the indeterminacy of common factors, the latent variables scores are not well determined. However, latent variable scores are well determined by PLS-SEM (Fornell and Bookstein, 1982; [Chin, 2010](#)).

3.2 Purposes of partial least squares structural equation modelling analysis in knowledge management

It should be noted here that “the purpose of PLS-SEM analysis” is more related to the nature of the purpose and its objectives than the advantages of the technique. Following Henseler’s suggestions (2018), we identify the following five purposes:

1. Confirmatory research understands the causal relationships between variables. In this case, the path model is built on the idea of testing causal hypotheses that specify how and why determined phenomena occur. When this happens there is a need for the confirmation of the model via both fit indices and global model verification.
2. Explanation purposes of PLS-SEM deal with the situation created for explaining a dependent variable. This analysis is based on explaining and maximising the explained variance of the dependent variable through the coefficient of determination (R^2) and the sign, size and signification of the path coefficients, which usually represent hypotheses that the model sets out to verify.
3. The exploratory research looks for a quick identification of potential relationships between variables. The exploratory implies an inductive form of reasoning, and therefore, PLS-SEM path models are fruit of a productive dialogue between the path model (theory) and data (reality). Indeed, several PLS path models are tested to find and explore what hide behind the data.
4. The descriptive studies are focussed on populations’ descriptions. Additionally, [Henseler \(2018\)](#) states that “PLS algorithm provides a prescription for dimension reduction”. Examples of this are diverse indices of customer satisfaction existent. That

description would apply to any latent variable based on a perfect combination of its indicators without error terms (i.e. composites) (Henseler, 2017). In this scenario, PLS-SEM algorithm permits to obtain true latent variables scores to describe dependent latent variables (composites).

5. The goal of predictive analytics is generating accurate predictions of new observations. These new observations can be considered as timely or transversal (i.e. predict observations not included in the original sample used to test the model). Therefore, the goal of this analysis is to provide timely prediction of new observations, both within the sample (via coefficient of determination) and outside the sample (via hold-out sampling) (Rigdon, 2016).

After reviewing the literature shown in Table II, there were only 33 KM studies declaring the purposes of their PLS-SEM analysis (52.3 per cent). Of the 33 PLS-SEM studies, 18 (54.5 per cent) were explanatory (i.e. testing hypothesis and maximise the variance explained of a dependent variable in a specified model); 8 were exploratory (i.e. dialogue between theory and data, exploring diverse PLS-SEM path models); 3 were confirmatory (i.e. finding model fit indices and confirm the global model as valid); 2 were predictive (i.e. focussed on the prediction ability of PLS-SEM); and finally, 2 were declared as predictive analysis but really were explanatory (i.e. they tested hypotheses). Consequently, from the 33 papers declaring PLS-SEM purposes, only 2 were wrong and the rest of PLS-SEM purpose declarations appear to be satisfactory.

3.3 Types of latent variables and measurement model specification

When we study PLS-SEM foundations, one statement emerges quickly, PLS-SEM is a multivariate data analysis technique for one type of latent variable named composite. According to Ali *et al.* (2017) and Rigdon *et al.* (2017), PLS-SEM can only use composites to achieve good estimations. Were researchers to use factors with PLS-SEM as estimation method, they would find bias in their research conclusions. This means that KM researchers using PLS-SEM should be clear with respect to the type of latent variable they used in their analyses.

After reviewing the type of latent variable researchers used in Table II, we found that only 33 KM papers declared the type of latent variable they used (52.38 per cent). Of these 33 papers, only 5 KM papers were right and they stated they were using composites in their PLS-SEM analyses. 26 works, however, told they were using factors when they were really using composites (78.78 per cent), and 2 papers told that they were using formative latent variables. Additionally, we review the estimation way of the composites in 63 PLS-SEM models in the KM domain. 47 studies of 63 were considered composites estimated on mode A (74.60 per cent), 9 were mixed models using mode A and B composites (14.28 per cent), and three PLS-SEM models only have composites estimated on mode B. Finally, four studies did not report the estimation way of their composites. According to these findings, we can see that there is a misunderstanding between KM researchers about the type of latent variable that should be used in PLS-SEM studies.

3.4 Assessment of global model fit

The assessment of the global model fit is a controversial topic that has recently received research attention (Henseler, 2018; Henseler *et al.*, 2016). It should also be noted that results from previous studies do not provide conclusive evidence for the incorporation of fit indices. In this vein, Henseler (2018) updated his own idea for fit indices and states that "it is only applicable for confirmatory studies".

Henseler *et al.* (2016) introduce the global model fit as starting point to model assessment in PLS-SEM. If the model does not adjust data, data possess more information than

specified model. In that case, these authors states that, obtained estimations could not make sense, and the conclusions can be questionable. The assessment of global model fit can be calculated through two ways: fit indices and inferential statistics ([Henseler et al, 2016](#); [Dijkstra and Henseler, 2015](#); [Henseler, 2017](#)). The first one is obtained directly from PLS-SEM algorithm and it is an approximated measure of fit and the second one is an exact measure of fit resulting from a bootstrapping procedure ([Henseler et al, 2016](#)). Additionally, another fit indice as the Goodness of Fit ([Tenenhaus et al, 2005](#)) was strong questioned by [Henseler and Sarstedt \(2013\)](#).

Thus, we reviewed if fit indices were used in confirmatory studies. Of the three confirmatory studies in KM papers sample, there was not any paper reporting fit indices. Independently of the paper purpose, some fit indices were reported in our KM papers sample. Regarding this, 12 papers (19.04 per cent) reported the Goodness of Fit (GOF) index developed by [Tenenhaus, Vinzi, Chatelin and Lauro \(2005\)](#) but questioned by [Henseler and Sarstedt \(2013\)](#), and 5 used the standardised root mean square residual (SRMR) as fit indices (9.52 per cent) and with respect to this index, it is the only that there is some type of agreement in the PLS-SEM community ([Hair et al., 2018](#)).

3.5 Assessment of measurement model

In ensuring and reviewing the measurement model, we use four core evaluating principles:

1. average variance extracted (AVE);
2. construct consistency;
3. use of Fornell and Larcker criterion and the heterotrait-monotrait (HTMT) ratio for discriminant validity; and
4. collinearity analysis for composites estimated in Mode B.

With regard to use of AVE, in the PLS-SEM reports in our papers sample, only 3 papers did not report the AVE value. The rest of them did report (95.23 per cent). According to the internal consistency, or construct consistency, it is important to note that there are three indices to assess it, such as: Cronbach's alpha, composite reliability and Dijkstra and Henseler's rho. Thus, all three indices were reported in five situations (7.93 per cent), then 33 models reported Cronbach's alpha and composite reliability (52.38 per cent), and only composite reliability was reported by 19 papers (30.15 per cent). Finally, Cronbach's alpha alone was reported 3 times and Dijkstra and Henseler's rho once only.

With regard to assessment of discriminant validity, there are currently two ways of assessment of discriminant validity: the well-known Fornell and Larcker's criterion and the most recent and accurate of HTMT criterion ([Henseler et al., 2016](#)). From 63 papers that validated the discriminant validity, all but 5 studies (92.6 per cent) used the Fornell and Larcker's criterion and 7 studies used HTMT criterion (11.1 per cent). It is worth to note that HTMT criterion was published in 2015 ([Henseler et al., 2016](#)), and therefore, these indices have not been adequately translated to the PLS-SEM before 2016. Future studies will consider increasingly the HTMT to assess the discriminant validity. Finally, of the 12 KM papers using composites estimated on mode B, 4 reported the variance inflation factor (FIV) and the other ones used criteria proposed for composites estimated on mode A. None of them used redundancy analyses and the same 4 reported the size and significance of the regression weights. These findings suggest KM researcher should consider the guidelines for assessing composites estimated on mode B ([Hair et al., 2017](#)).

3.6 Assessment of the structural model

After evaluation of measurement model, the assessment of structural model follows. The first step to this assessment – because of the type of algorithm that uses PLS-SEM for its

estimations (i.e. ordinary least squares, OLS) – is the analysis of the collinearity of the exogenous construct of a determined endogenous construct. Hair *et al.* guidelines (2017) suggest that there is a collinearity suspect when FIV values are higher than five and tolerance levels greater than 0.20. In our review, we can find 16 studies (25.39 per cent) analysing collinearity in the structural model. This is an important issue to be tested because high levels of collinearity generate estimation bias in the coefficients calculated in the structural model.

Another important fact to consider is how the significance test for path coefficients was performed. Currently, this hypothesis test (i.e. significance test) is performed depending on the sign of the hypothesis. If one hypothesis makes a statement about the relationship sign (i.e. positive or negative), a one-tailed test should be performed. In the turn, if one hypothesis did not determine a specific sign, a two-tailed test would be right. Independently, if the significance test is related to a student distribution (*t*-value) or a confidence interval (e.g. percentile), the rule of tails is applicable (Hair *et al.*, 2017).

In our review, we checked if researcher reported tails in their significance test (i.e. hypothesis test) and if they reported it well. However, only 21 out of a total number of 63 PLS-SEM testing hypotheses reported explicitly the tails used in their significance test (33.33 per cent). In addition, 11 papers out of 21 PLS-SEM reporting tails used a two-tailed test when they should have used one-tailed. Therefore, only 10 PLS-SEM studies tested correctly the significance test for their hypotheses. This means researcher do not know clearly the rule of tails for their hypotheses' test.

Additionally, as PLS-SEM is a non-parametric test, standard parametric significance tests are no very adequate to show if the path coefficients is significantly different from 0 (Hair *et al.*, 2014). For this reason, PLS-SEM uses the bootstrapping procedure to derive a bootstrap confidence interval with a certain level of confidence (Chin, 1998). After our review, only 7 out of a total number of 63 PLS-SEM studies testing hypotheses reported the confidence interval for the significance of path coefficients (11.11 per cent). As new guidelines to report PLS-SEM studies consider interval confidence to test the significance of path coefficients (Hair *et al.*, 2017, 2018; Henseler *et al.*, 2016), KM researcher should use confidence intervals for either supporting or not supporting hypotheses.

3.7 Use of predictive analysis

Although the predictive approach is one of the purposes that can be covered by PLS-SEM, only 2 out of a total number of 63 chose this purpose. This is an important limitation, given the characteristics of PLS-SEM analysis. In fact, all the studies have a predictive face and the prediction of the model should be specially reported in studies with predictive goals. In doing so, the Stone–Geisser's test for predictive relevance (Q^2) has been usually used by researchers as criterion to measure the predictive relevance of the dependent constructs estimated on Mode A. It should be noted that this test has been questioned by Shmueli *et al.* (2016), who suggest that the Q^2 obtained by their procedure is preferable where possible. Based on our review, 18 PLS-SEM studies out of a total number of 63 studies used Stone-Geisser's test (Q^2) (28.57 per cent).

3.8 Use of a mediation analysis

Out of 63 studies revised and reviewed, 20 performed a mediation analysis (31.75 per cent). Furthermore, ten of these papers still used the Baron and Kenny's approach (1986) and the Sobel's test. The remaining ones calculated and reported very well their mediation analysis following the instructions from Nitzi *et al.* (2016).

3.9 The software package to perform partial least squares structural equation modelling estimations

In our KM papers' review, we noted that while 17 PLS-SEM did not report any software (26.98 per cent), whereas 15 papers did it without any citations (23.81 per cent). The software packages cited in PLS-SEM studies in KM field were: SmartPLS 33 times (52.38 per cent), 11 of them were not cited; the plspm R code once, ADANCO was used 2 times, PLS-Graph 6 times, and WrapPLS was used in 5 occasions. As a result, the most used software currently is SmartPLS ([Ringle et al., 2005](#)). In sum, it is very important reporting the software that helps to carry out PLS-SEM and citing accordingly.

4. Discussion

Broadly speaking, KM researchers have operationalised knowledge structures by the adoption of enablers versus outputs. While the utilisation of enablers may show how and to what extent indicators have to be combined to shape the construct itself, the presence of outputs may show how and to what extent the indicators happen as a result of the latent variable. From the point of view of KM researchers, it is also important to differentiate between factor and composites. On this regard, [Rigdon \(2016\)](#) provides nonetheless an interesting perspective about factors and composites measurement. In a multiple indicator model to treat his constructs as factors or composites based on the conceptual variables that they represent, he represents a conceptual variable like "satisfaction" being "formed out of" a set of observed variables, or we think of a common factor as "causing" variation in a set of observed variables. He concludes that both notions are absurd. In the first case, the set of observed variables form a representation or image of the conceptual variable, not the conceptual variable itself. In the second case, variation in the indicators is caused by actual conceptual variables (almost certainly more than one), while the factor is only an image with no causal power.

In view of the foregoing considerations, KM academics ought perhaps to be aware that many of the measures they are using are more composites than factors. Taking into account the widely differing purposes of PLS-SEM analysis, when operationalising KM variables that are called "artificial unobserved variables", researchers should not impose any restrictions on covariances between indicators of the same construct ([Henseler, 2017](#)). This means that although PLS-SEM is popular among many researchers in KM, in investigating this technique, however, this study identified several common mistakes that raised some concerns about the use of this method among KM researches. In view of the foregoing considerations, there are four areas into which the main problems of KM researchers can be classified:

1. reasons and purposes of PLS-SEM study;
2. determination and operationalisation of the latent variables;
3. structural model evaluation; and
4. reporting results and analysing using PLS-SEM.

This study tries to find realistic solutions to these issues below.

It is important to start reporting *the reasons and purposes of PLS-SEM*. The reason why to decide to use PLS-SEM for data analysis technique should be based on the type of latent variables that is used to measure non-observable variables as proxies of the theoretical concept of research ([Rigdon, 2016](#)). Additionally, and complementary, one should state what kind of purposes has with the PLS-SEM analysis (e.g. descriptive, exploratory, explanatory, confirmatory or predictive). This statement determines the content of the analysis and of the report that consequently will be come out ([Henseler, 2018](#)).

Regarding *the determination and operationalisation of the latent variables*, it seems reasonable to think that the large majority of constructs included in [Table I](#) are either composite reflective constructs (mode A estimation) or composite formative constructs (mode B estimation). Taking in account the type of use of given to latent variables and the fact that composites and factors have often been used interchangeably throughout the published KM literature, KM researchers should use consistent PLS and/or CBSEM ([Dijkstra and Henseler, 2015](#)), which allow them to resemble a factor model ([Ali et al., 2017](#)), and PLS-SEM if they are using composites as latent variables.

It is important to note that in the case of composite reflective constructs (mode A estimation), the Dijkstra and Henseler's rho (pA) and the average variance extracted (AVE) should be above the common standards of 0.7 and 0.5 respectively ([Dijkstra and Henseler, 2015](#); [Henseler et al., 2016](#)). In addition, all composite regression weights from composite formative (mode B estimation) constructs have to be non-collinear and most of them relevance and statistically significant ([Hair et al., 2017](#)). With regard to composite formative constructs, weights also measure the relative contribution of each indicator to its construct (ordinal range) ([Hair et al., 2017](#)).

Another important issue relates to the fit statistics for the models, and while in the past the priority was the predictive relevance of the model in determining a given dependent variable, today PLS-SEM offers fit measures such as SRMR, d_ULS, d_G or NFI ([Henseler et al., 2016](#)). In this vein, the SRMR value of the measurement model should be lower than 0.08 and the Bentler-Bonett index or normed fit index greater than 0.90 ([Hair et al., 2017](#)). In addition, all discrepancies (i.e. d_ULS, d_G) should be below the 99 per cent quantile of the bootstrap discrepancies (Hl₉₉), which may suggest a very good measurement model fit ([Henseler et al., 2014](#)).

With regard the *structural model evaluation*, an important issue relates to the structural/inner model, bootstrap confidence intervals constitute a good approach for detecting path coefficients ([Hayes and Scharkow, 2013](#)). In this vein and based on [Streukens and Leroi-Werelds \(2016\)](#), the use of bootstrapping (e.g. 5,000 or 10,000 resamples) produces bootstrap confidence intervals of standardised regression coefficients. Additionally, when KM researchers perform their hypothesis or significance test they should consider and report the number of tails in their test. In this paper, we have detected most of KM researchers forgot it and sometimes made wrong selections with reference to the test they wanted to perform. It is important to note a significance test will use one-tail whether one direction or sign has been determined or hypothesised and two tails when no direction or no sign has been stated.

Another important issue is about problems of *reporting results and analysing using PLS-SEM*. On the basis of current literature, [Ali et al. \(2017\)](#) proposed to consider reporting bootstrap confidence intervals, rather than only using *p-value*. In addition, the best test to assess discriminant validity is the HTMT ratio ([Sarstedt et al., 2016b](#)), and it is preferable to the two traditional methods for discriminant validity assessment, such as cross loadings and the Fornell-Larcker criterion ([Fornell and Larcker, 1981](#)). Another interesting point here is the suggestion that KM researchers to run a mediation analysis should compare direct and indirect effects using bootstrapping ([Nitzl et al., 2016](#)), rather than using [Baron and Kenny's \(1986\)](#) method in partnership with the Sobel test ([Koopman et al., 2014](#)). Finally, as a result of PLS-SEM developments in recent years, and as the Stone-Geiser's test (Q^2) does not consider the entire rows of the dataset (i.e. there is no full observation omission), it is highly recommended to exchange the Stone-Geiser's test (Q^2) to the Q^2 developed by [Shmueli et al. \(2016\)](#).

Bearing above ideas in mind and based on [Haans et al. \(2016\)](#)'s work, [Table III](#) provides a few examples of research questions to different reasons to use PLS-SEM. In addition, [Table III](#) shows a workflow of steps or checklist to follow by the researcher that wants to use PLS-SEM.

Table III Workflow of steps to follow by the researcher that wants to use PLS-SEM

Determine if reasons to use PLS-SEM are adequate

Type of latent variables

Complexity of model

Necessity of using latent variable scores

Determine PLS-SEM purposes

Confirmatory: Utilisation of global model fit indices

Explanatory: Maximising of explained variance of dependent variable(s) and sign, magnitude and significance of path coefficients

Explorative: Development of several models looking for how are the relationships between latent variables. Dialogue between theory and data.

Descriptive: Using the estimation of latent variable scores to describe the dependent latent variable

Predictive: Determine if our model with our data predict other data (time, context, industry...)

Assessment of our path model aligned with our purposes

Specification of the path model

Determination and operationalisation of the latent variables

Step 0: Evaluation of global model fit (only for confirmatory purposes)

SRMR, d_ULS, d_G or NFI (estimated model)

Step 1: Evaluation of the Measurement Model

SRMR, d_ULS, d_G or NFI (saturated model)

Step 1a:

Composites estimated on mode A

Internal Consistency

Construct Validity

Discriminant Validity

Step 1b:

Composites estimated on mode B

Collinearity among Indicators

Significance and Relevance of Outer weights

Step 2: Evaluation of Structural Model

Collinearity Assessment

Sign, magnitude and significance of Structural Model Path Coefficients and Hypothesis Testing

Coefficient of Determination (R^2)

Effect size f^2

Issues reporting PLS-SEM results

Report bootstrap confidence intervals

Report HTMT ratio

Significance of the indirect effects in mediation analysis

Q2 developed by [Shmueli et al. \(2016\)](#) (predictive purposes) Report software used

Source: Own elaboration adapted of [Samani \(2016\)](#), [Hair et al. \(2017\)](#) and [Henseler et al. \(2016\)](#)

It is important to note here that during the course of our research, we faced several general limitations of methodological research. Inevitably, the quality of the analysis heavily depends on the quantity and quality of studies reviewed. Therefore, further research needs to be conducted to investigate other journals (e.g. the learning organisation or management decision) which have not been included in this study but are also likely to publish papers related to KM. Secondly, although this paper presents common mistakes which occur frequently among PLS users, the mistakes presented are not exhaustive and assumes that other errors may affect the outcomes and enablers when using the PLS-SEM analysis. Finally, despite an extensive literature review of the past three years, it is likely that a more extensive review would pay significant dividends in extending the richness and breadth of the mistakes presented. Nevertheless, we feel comfortable in having done our best to present typical mistakes when using the PLS-SEM analysis and hope that this work will support KM researchers.

5. Conclusion

This study contributes to the vast KM literature by documenting the problems and misconceptions related to PLS-SEM. Findings show that KM researchers have not fully

exploited PLS-SEM potential regarding advanced analysis techniques. There were two problems arising from the purposes of PLS-SEM analysis. First, researchers must declare the purposes of the PLS-SEM mainly because the report is different depending on the purpose of PLS-SEM analysis. Second, researchers must declare one purpose and carry out analysis and a report consistent with such a purpose. For example, as the report of exploratory analysis can be almost similar to explanatory, and even, confirmatory, KM researchers should offer a different interpretation. With a confirmatory and/or explanatory study, there is one model, one research design, one data set, one analysis, and a set of results and, therefore the uncertainty is captured only in the standard errors. However, in an exploratory situation this uncertainty may be linked to different aspects such as two or more models, different research design, multiple model test, and so on. Determining specific guidelines for each PLS-SEM purpose is an issue still uncovered, mainly for exploratory and descriptive PLS-SEM studies.

As KM researchers are looking for a way to operationalise unobserved and latent measures that are not directly observed, often data simply does not exist and only after a survey and using PLS-SEM data can be operationalised. It is also important to remark when PLS-SEM is not a suitable tool to data analysis in KM field. KM researchers should not use PLS-SEM, when their variables are not composite; when they measure all their theoretical concept with a single item instead of multiple items; and their dependent variables are categorical or nominal (Hair et al, 2017). Additionally, if they are specifying only common factors for latent variables, the best option is using CBSEM. If the path model contains a mix of common factor and composites, the best option is PLSc.

KM researchers should stress the purposes of their PLS-SEM analysis to facilitate papers and reports reading. As was commented the PLS-SEM analysis and reporting is conditioned by the PLS-SEM purposes. Therefore, it is important to report the purpose of PLS-SEM analysis to understand the research that was performed. This means that KM researchers need to improve their analyses and complement their results. KM researchers should, for example, look at how their findings can be affected by unobserved heterogeneity, levels of internal consistency and the degree of common method variance. It should be noted here that there are other problems to overcome, particularly those of endogeneity, longitudinal data analysis, multilevel modelling, and categorical latent variables as endogenous (i.e. dependent) construct. Although these advances are still in their early stages of development, future KM studies should exploit and, if possible, further develop these methods.

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